Laporan Praktikum Instalasi Debian OS dan OpenSUSE di VirtualBox



Kelompok 2 :

2504130089 BIMA ADHI PRATAMA KHARIS

2504130127 ALIF SATYA WICAKSONO

2504130050 ZAKKA NOVIATAMA

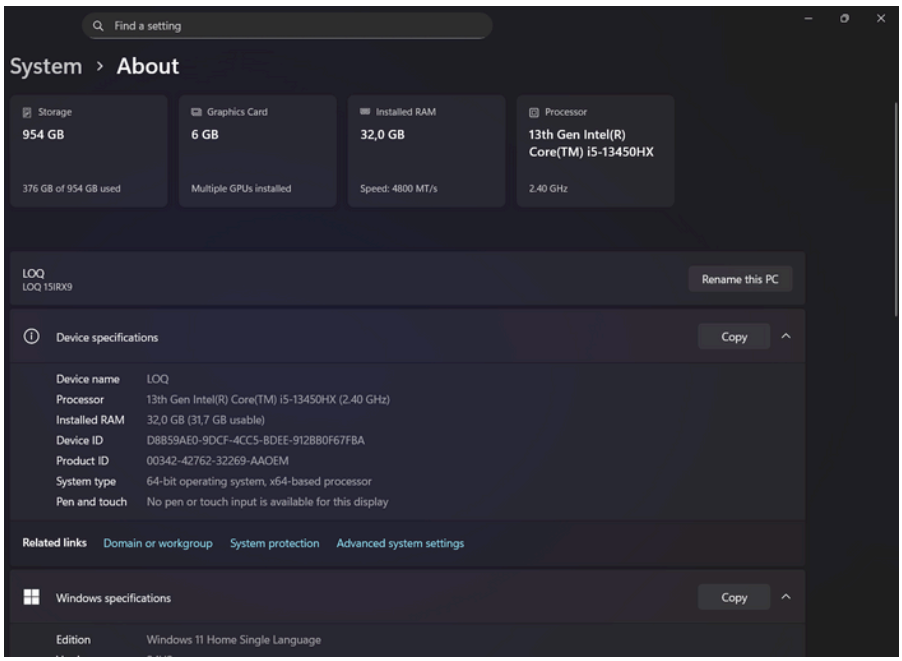
2504130106 MUHAMMAD DZAKY ALTHAF

PROGRAM STUDI TEKNIK INFORMATIKA FAKULTAS MATEMATIKA DAN ILMU PENGETAHUAN ALAM
UNIVERSITAS NEGERI SEMARANG 2025

Tahapan Instalasi Debian OS 13.1.0 XFCE VirtualBox 2GB/10GB

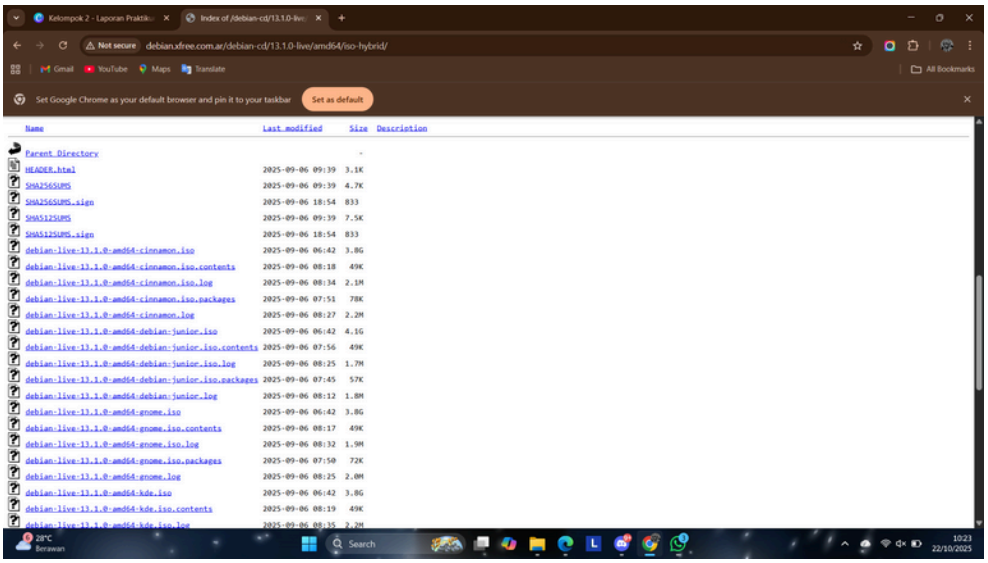
1. Persiapan

1. A. Spesifikasi laptop/PC memadai (RAM minimal 4 GB, storage kosong 15 GB).

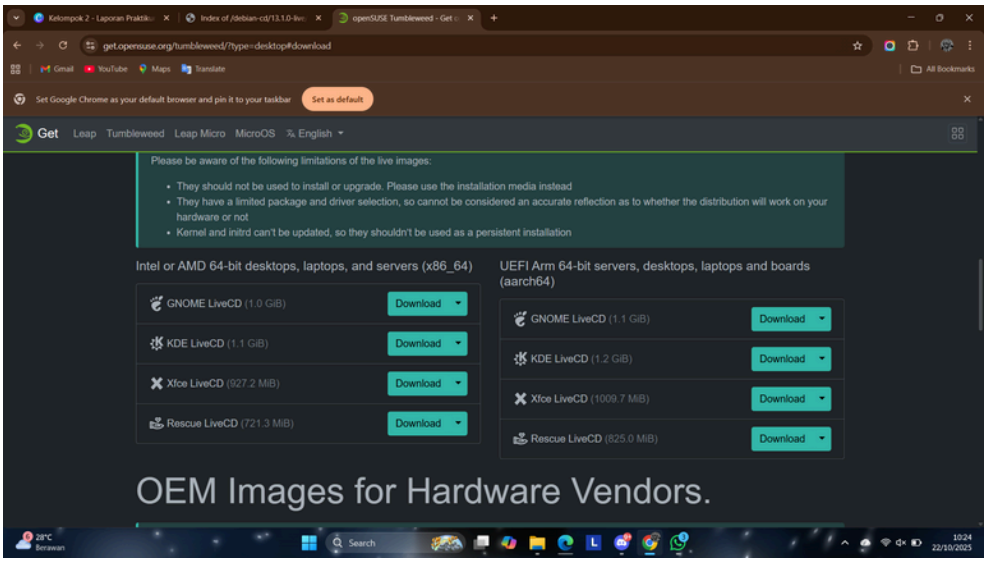


2. Download File ISO Debian dan OpenSUSE

2. A. File ISO Debian

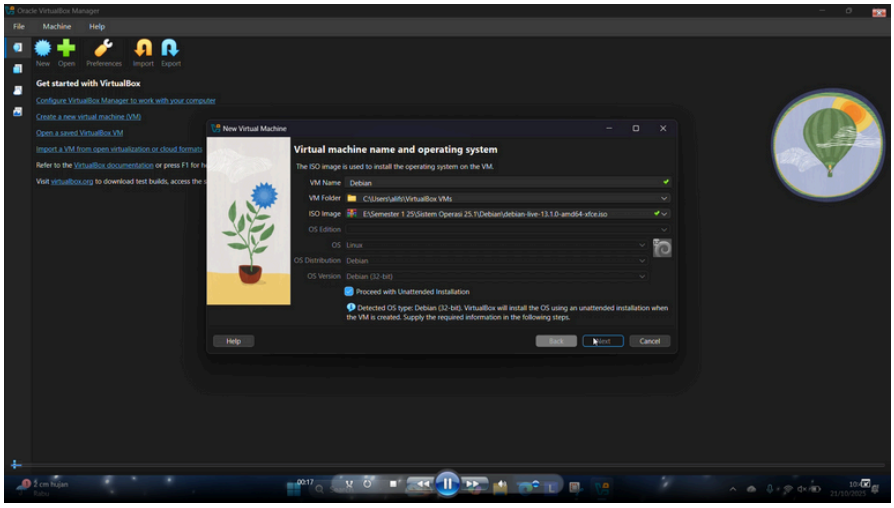


2. B. File ISO OpenSUSE

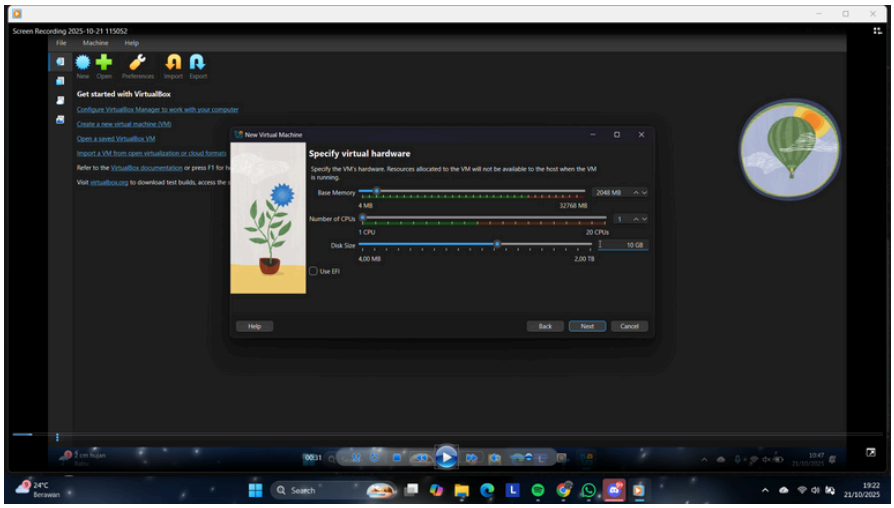


RAM 2GB / DISK 10GB
3. Membuat virtual machine Debian

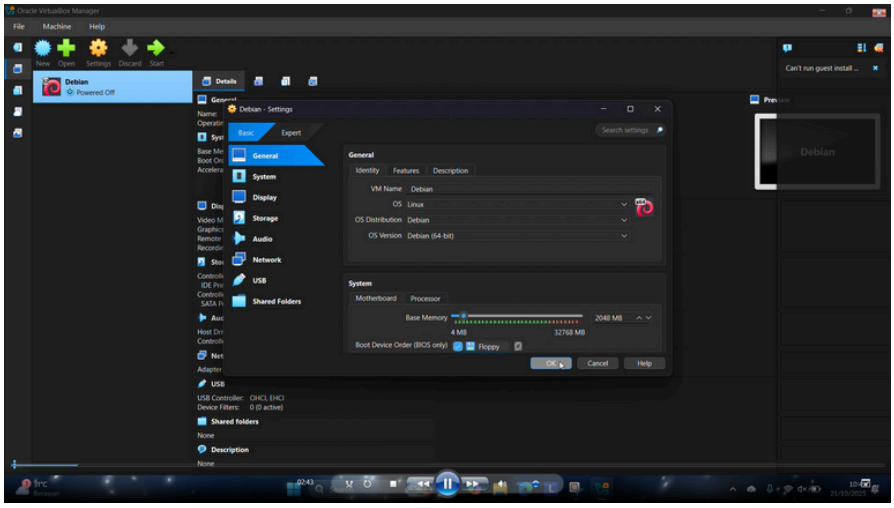
3. A. Masukkan file ISO kemudian pencet next



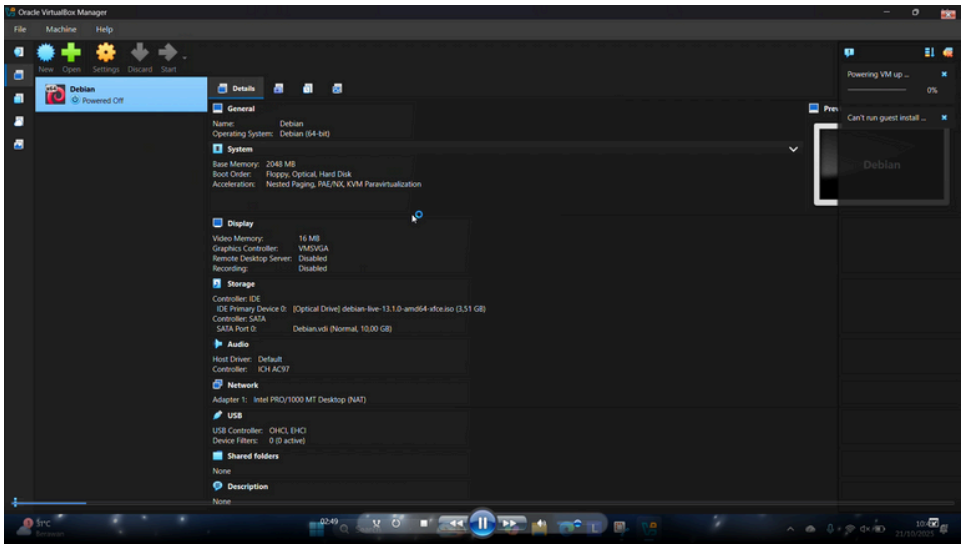
3. B. Atur RAM ke 2GB, CPU 1, dan Disk Size 10GB



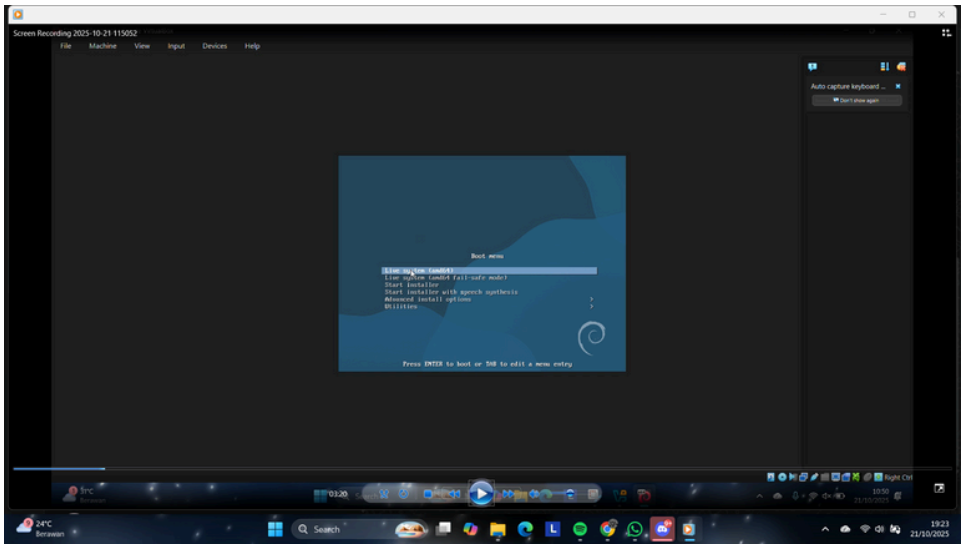
3. C. Ubah OS Version ke 64-Bit



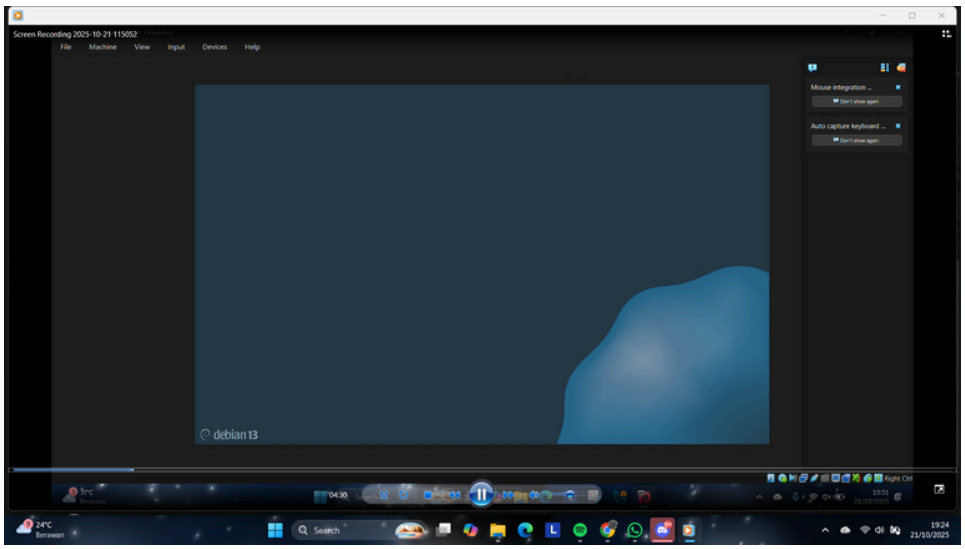
3. D. Start



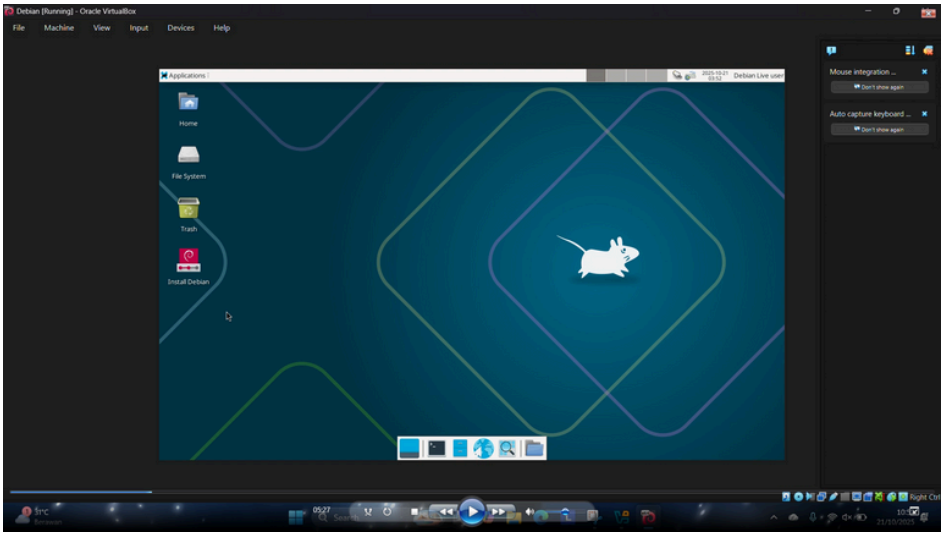
3. E. Pilih Live system (amd64)



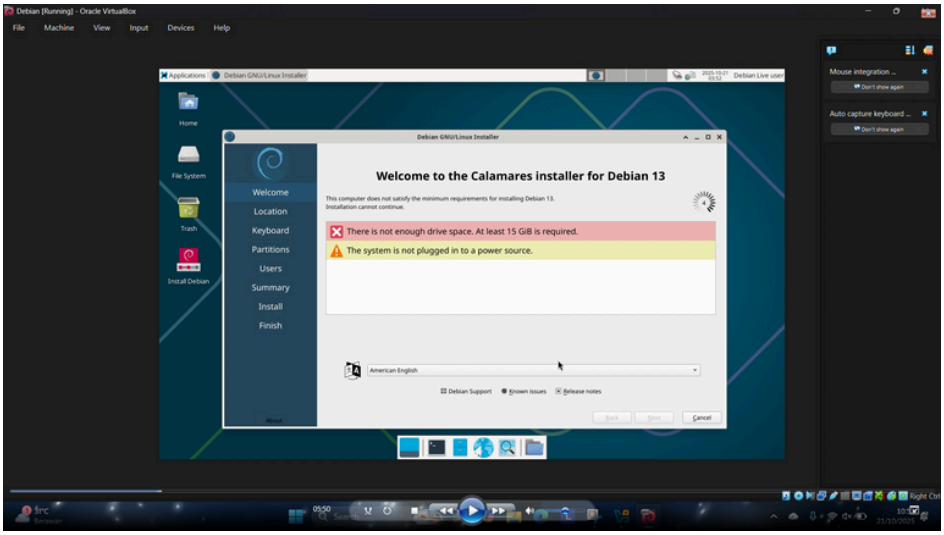
3. F. Tunggu proses booting



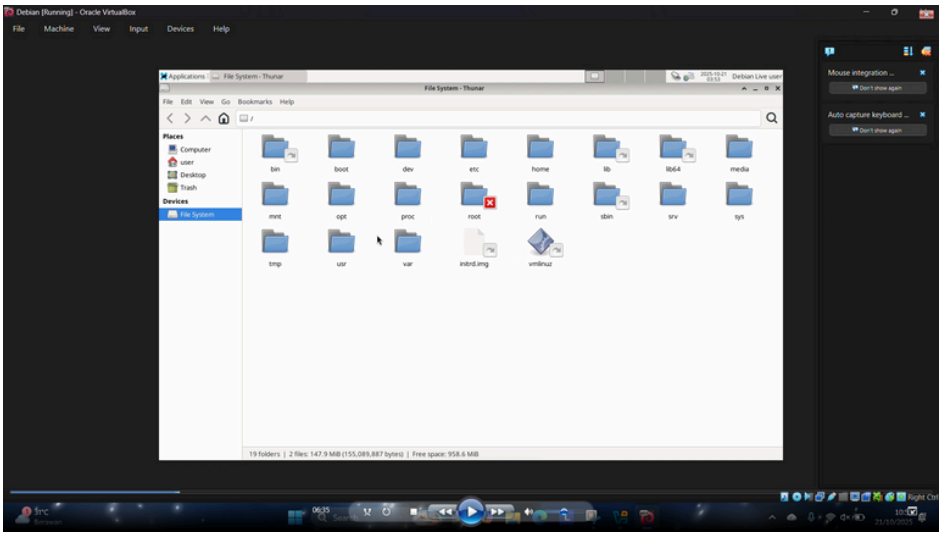
3. G. Tampilan awal Debian OS



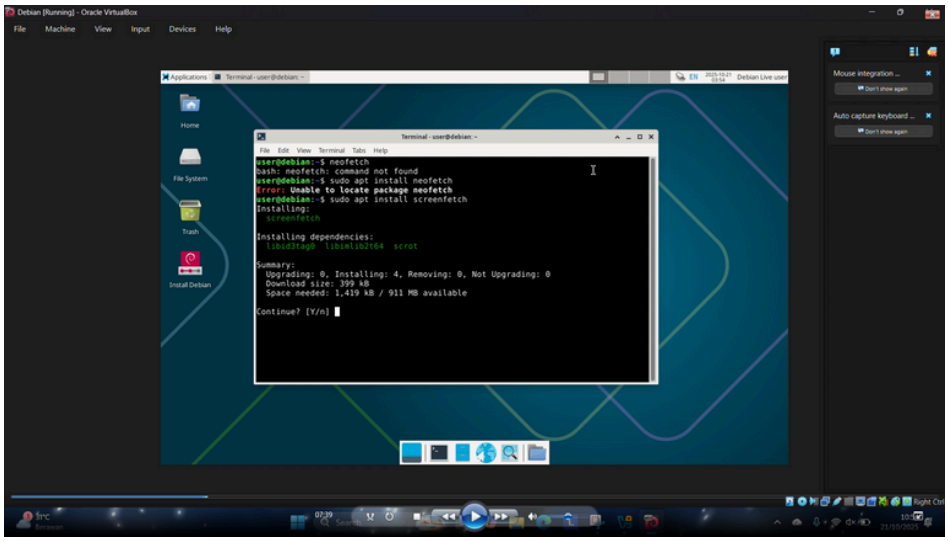
3. H. Klik icon install Debian OS → Tidak dapat melanjutkan proses instalasi karena minimal storage 15 GB



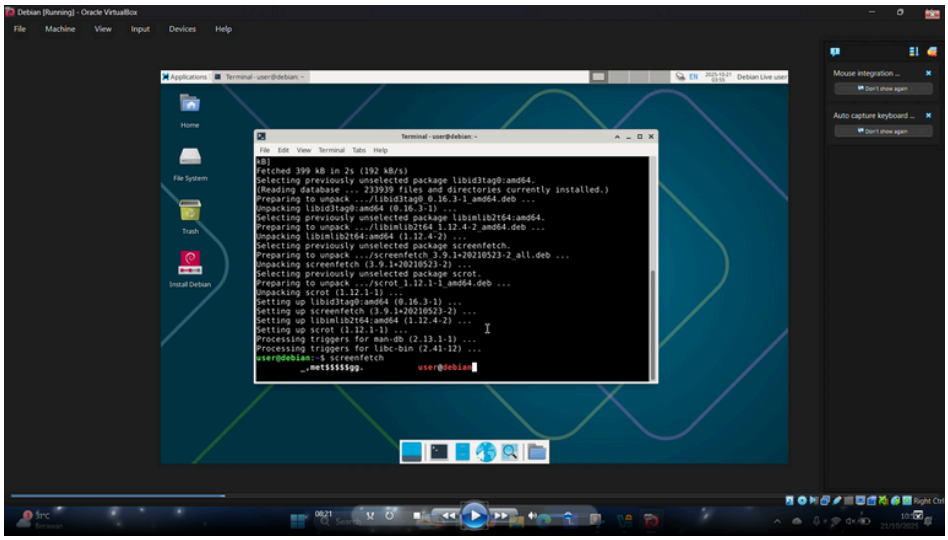
3. I. Sisa Storage adalah 948,6 MB dan tidak dapat melakukan instalasi



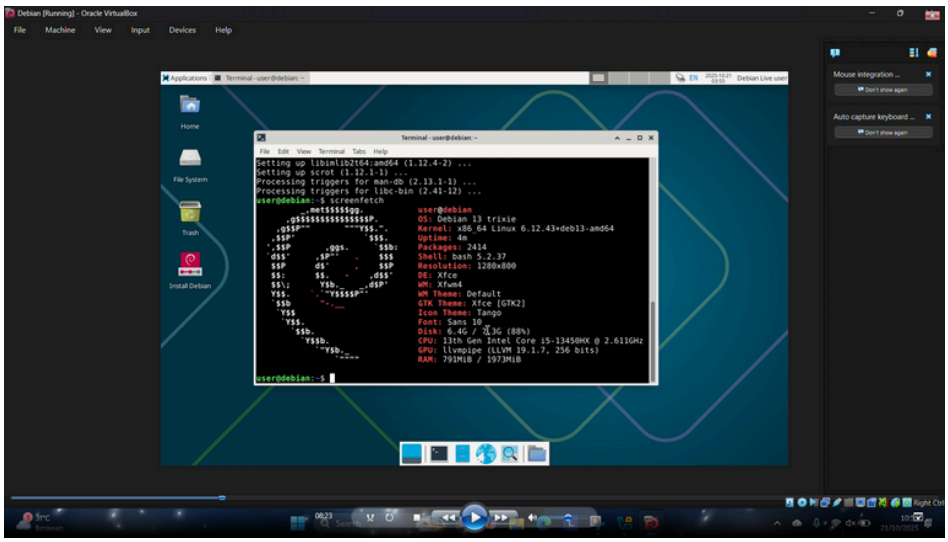
3. J. Ketik “Sudo apt install screenfetch” kemudian “y”



3. K. Ketik “screenfetch”

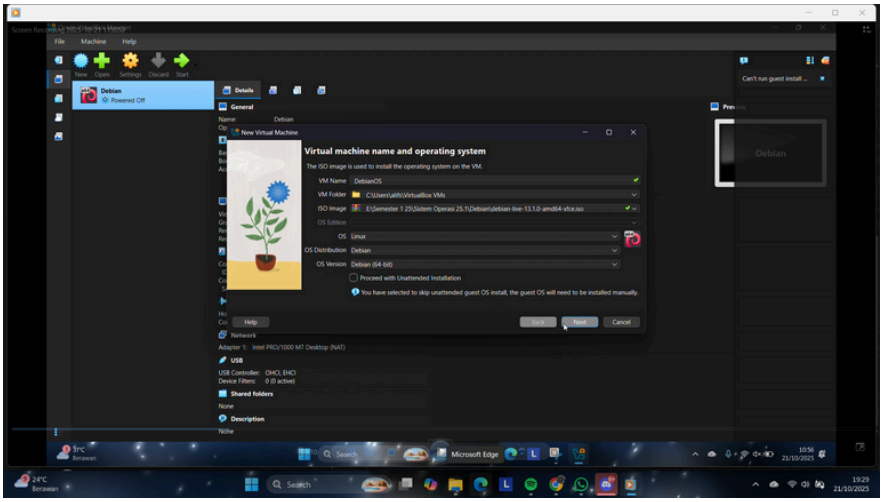


3. L. Pemakaian RAM 791MiB / 1973MiB

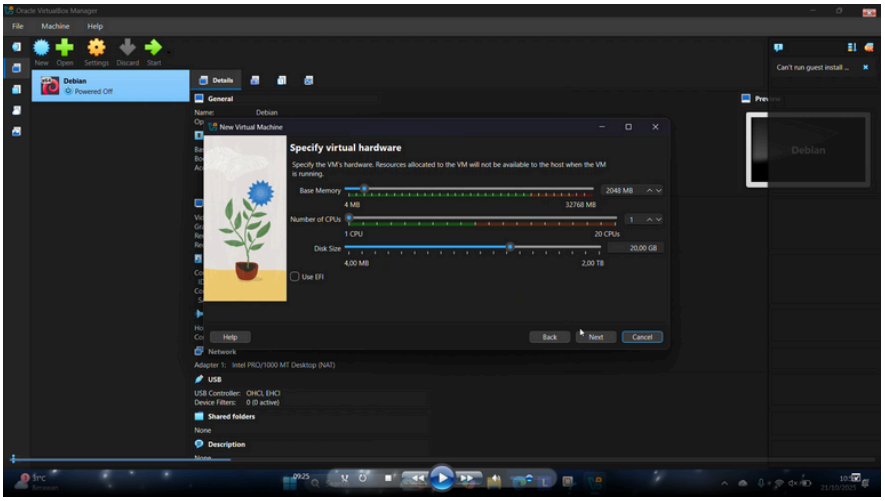


RAM 2GB / DISK 20GB
4. Membuat virtual machine Debian

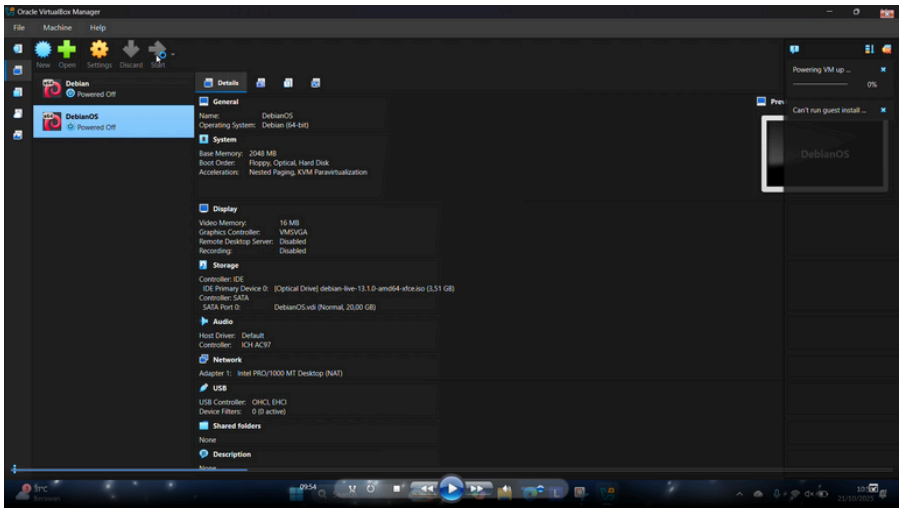
4. A. Masukkan file ISO kemudian pencet next



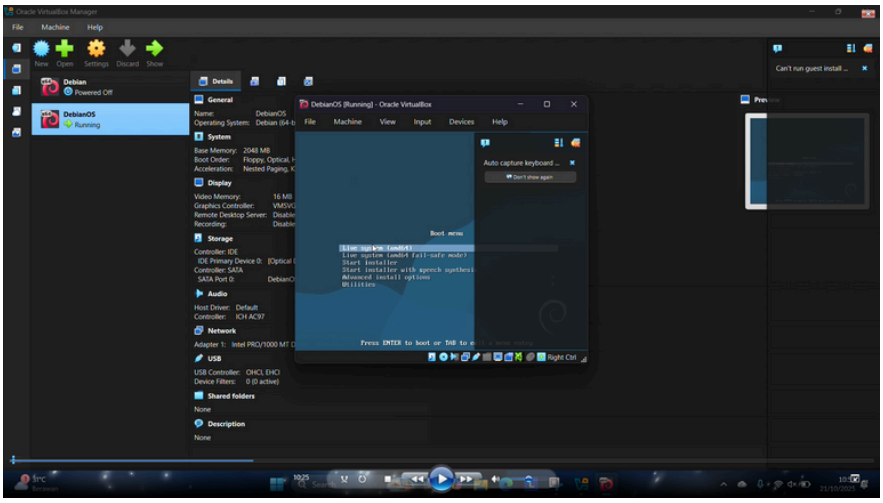
4. B. Atur RAM ke 2GB, CPU 1, dan Disk Size 20GB



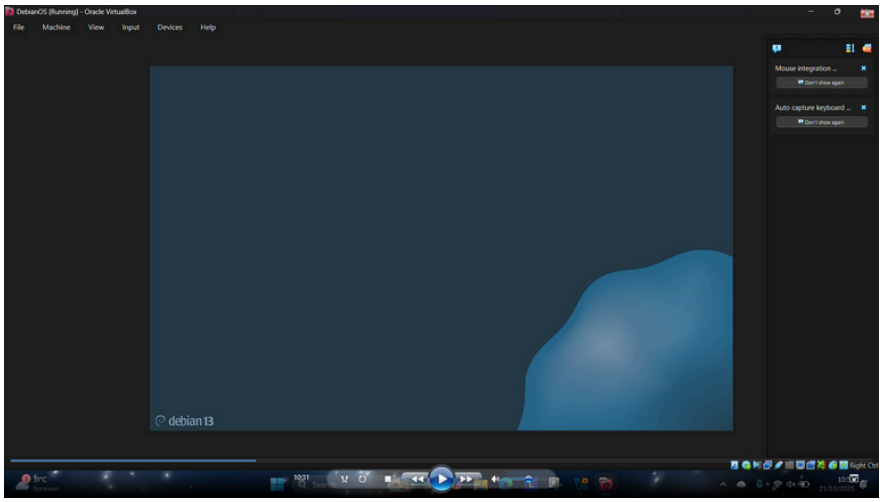
4. C. Klik start → Tunggu proses VM Powering



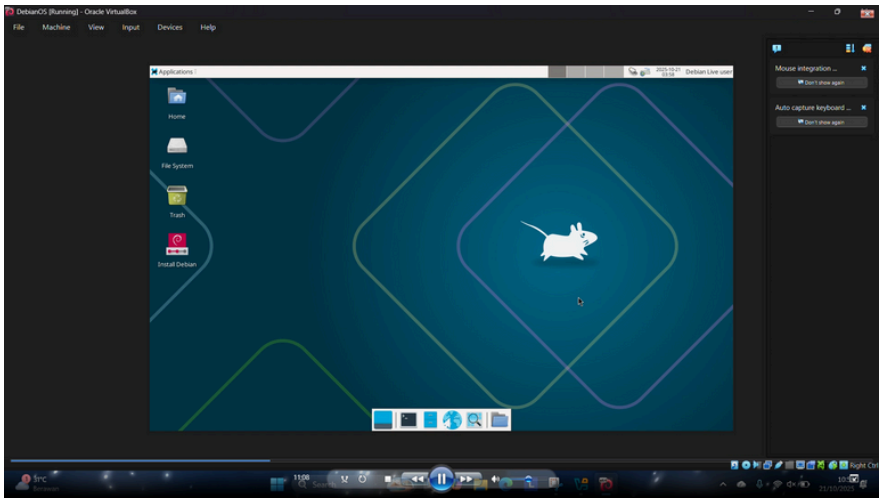
4. D. Pilih live system (amd64)



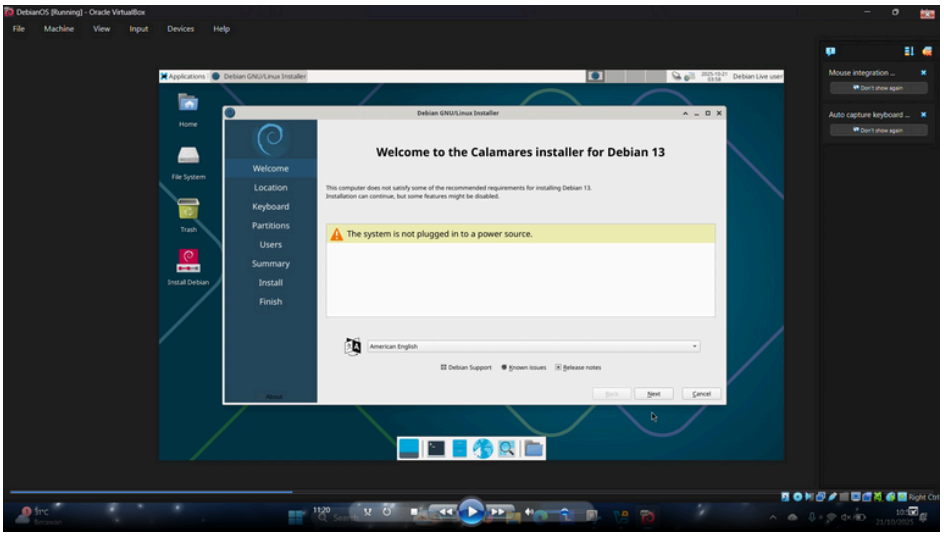
4. E. Tunggu proses booting



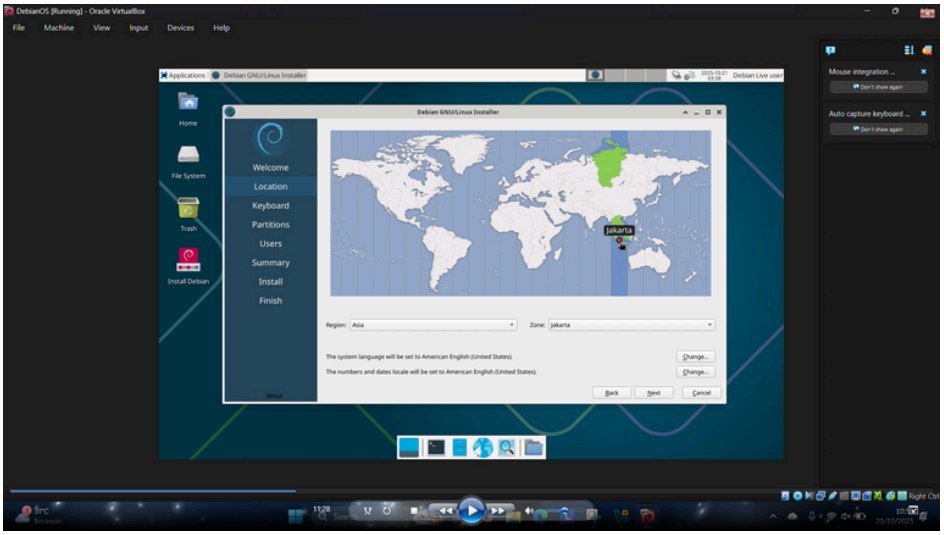
4. F. Tampilan awal Debian OS



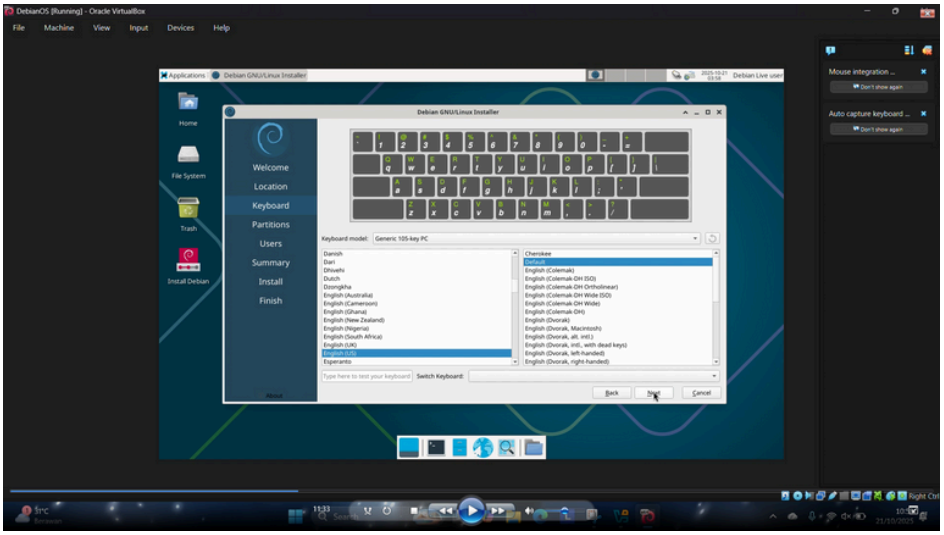
4. G. Klik instal Debian → klik next untuk melakukan instalasi



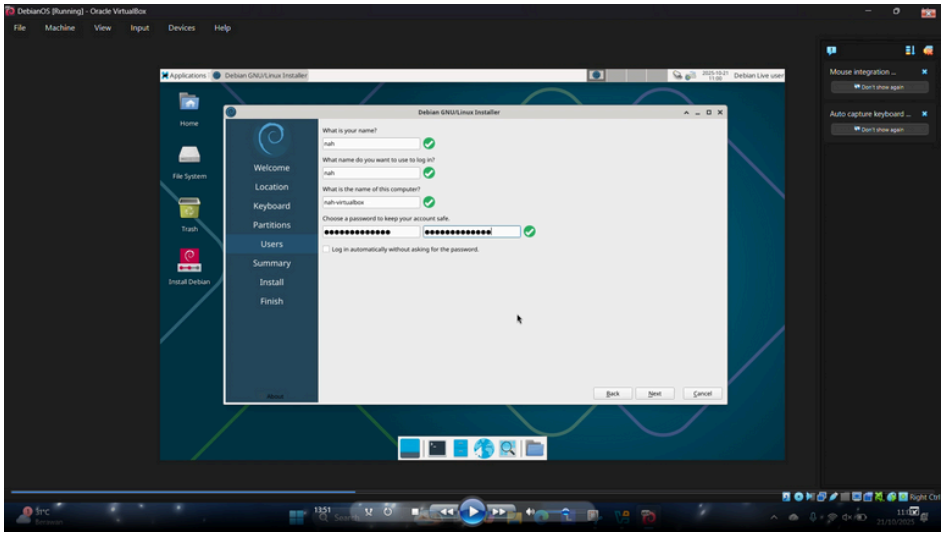
4. H. Pilih wilayah



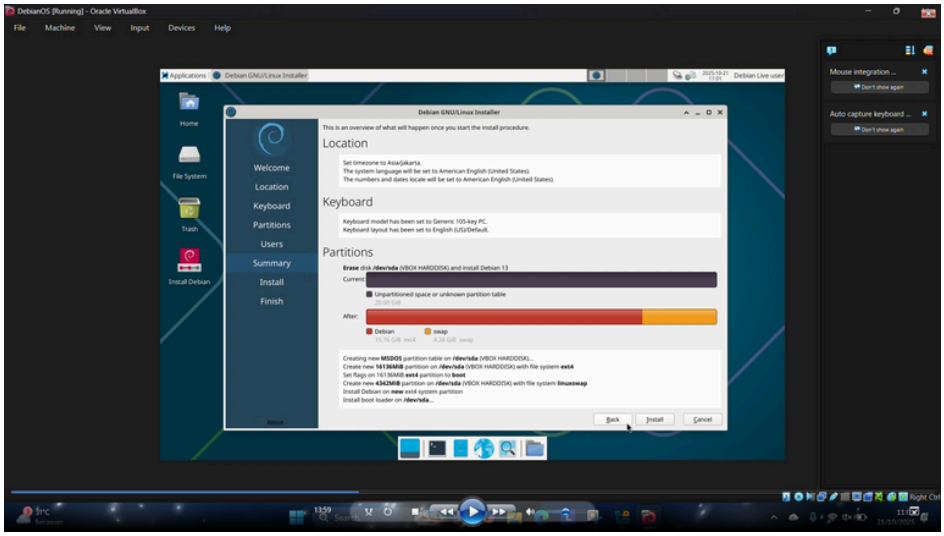
4. I. Pilih setup keyboard yang akan digunakan



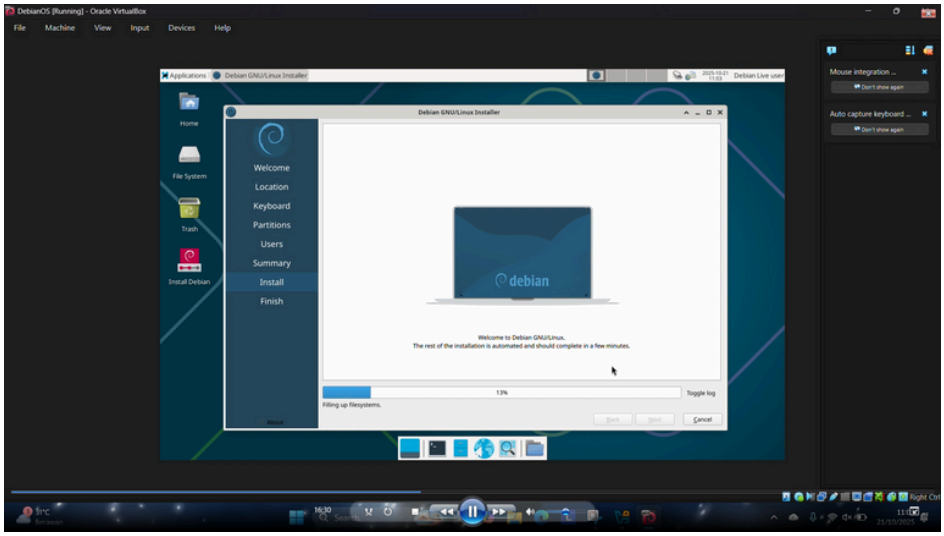
4. J. Masukkan nama dan password untuk PC/Laptop tersebut



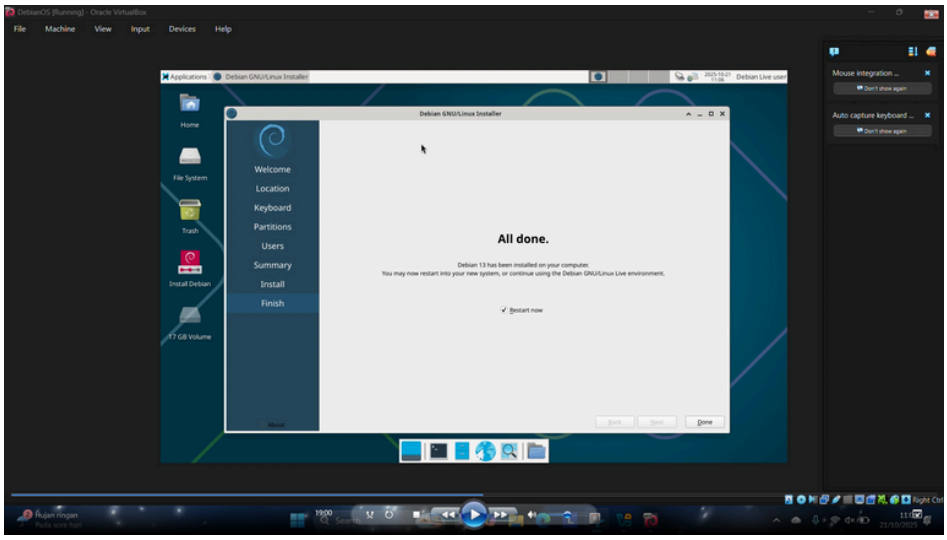
4. K. Cek kembali semuanya → Klik next jika sudah



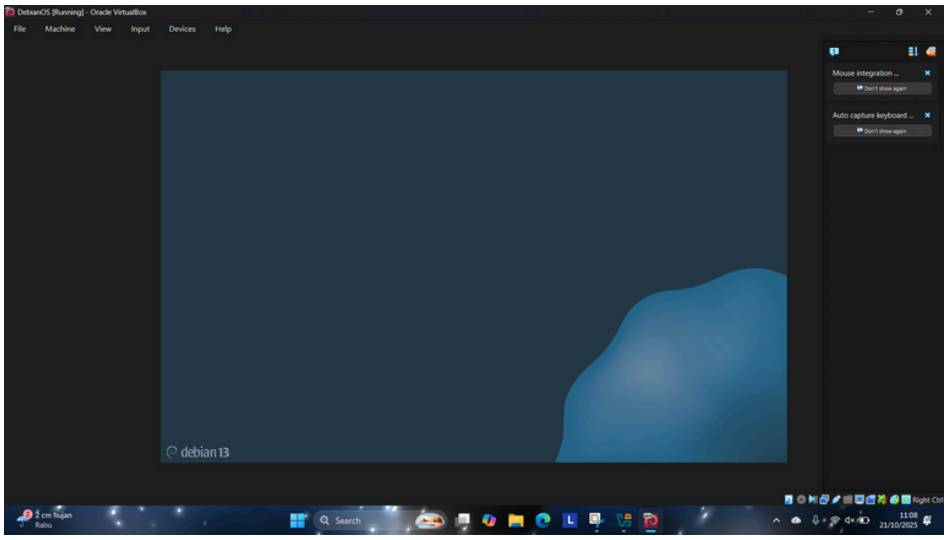
4. L. Tunggu proses instalasi Debian OS



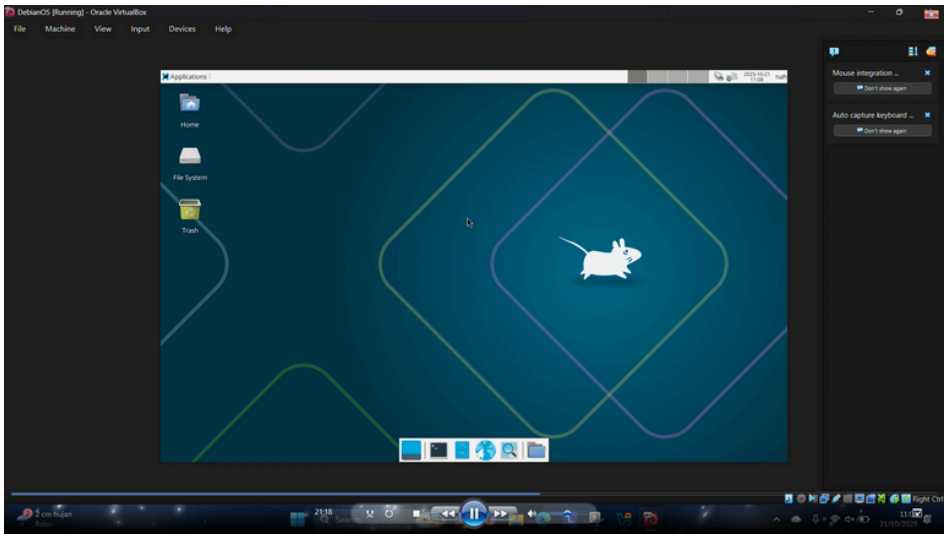
4. M. Klik Done jika sudah selesai melakukan instalasi untuk melakukan rebooting



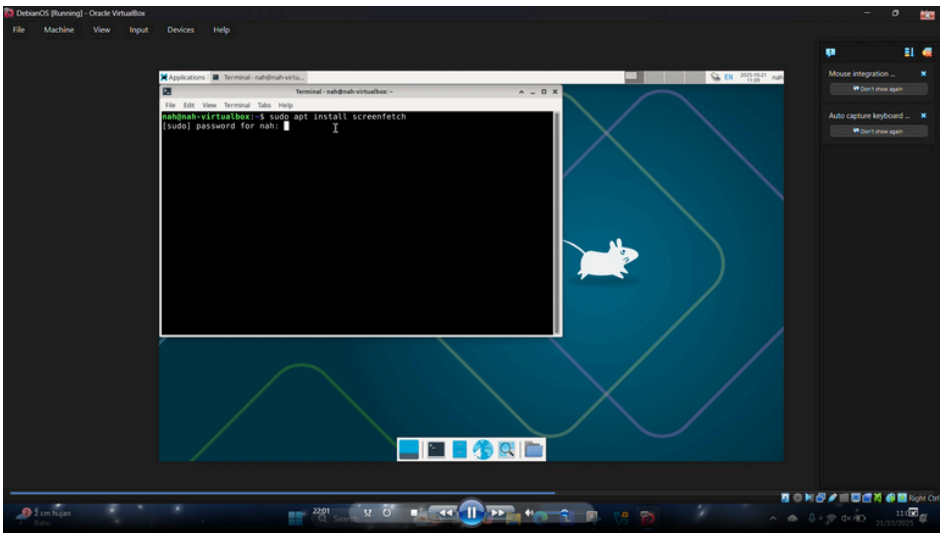
4. N. Tunggu proses booting kembali



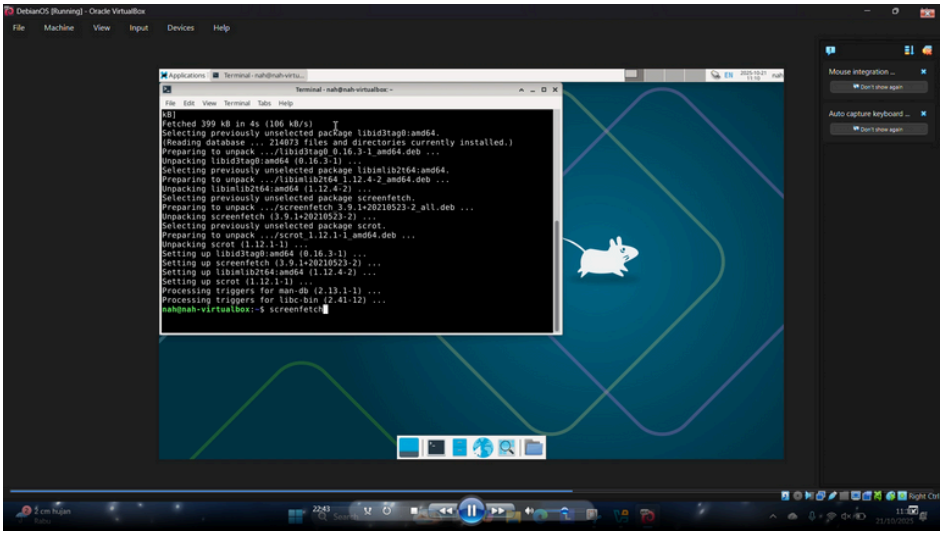
4. O. Tampilan awal setelah proses instalasi



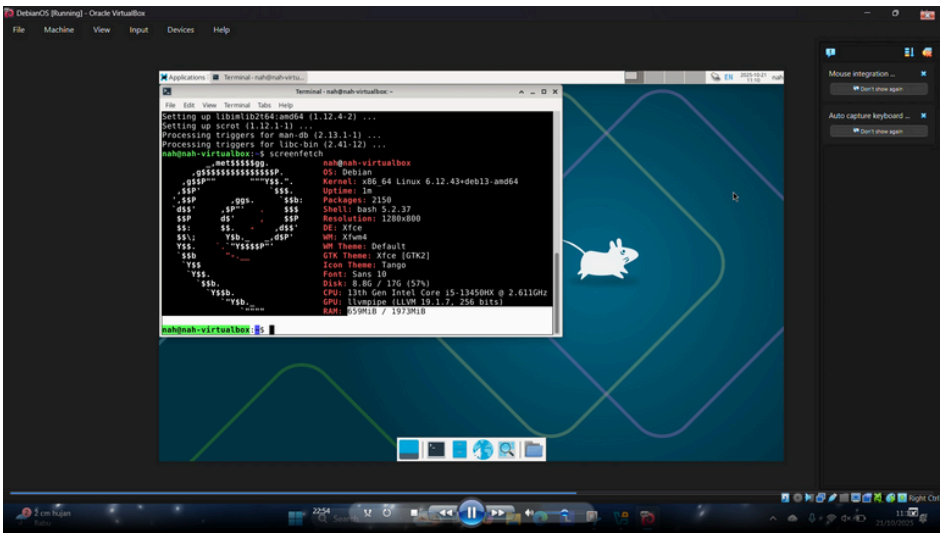
4. P. Masuk ke dalam terminal → ketik sudo apt install screenfetch



4. Q tunggu proses instalasi → Ketiksn screenfetch pada terminal tersebut



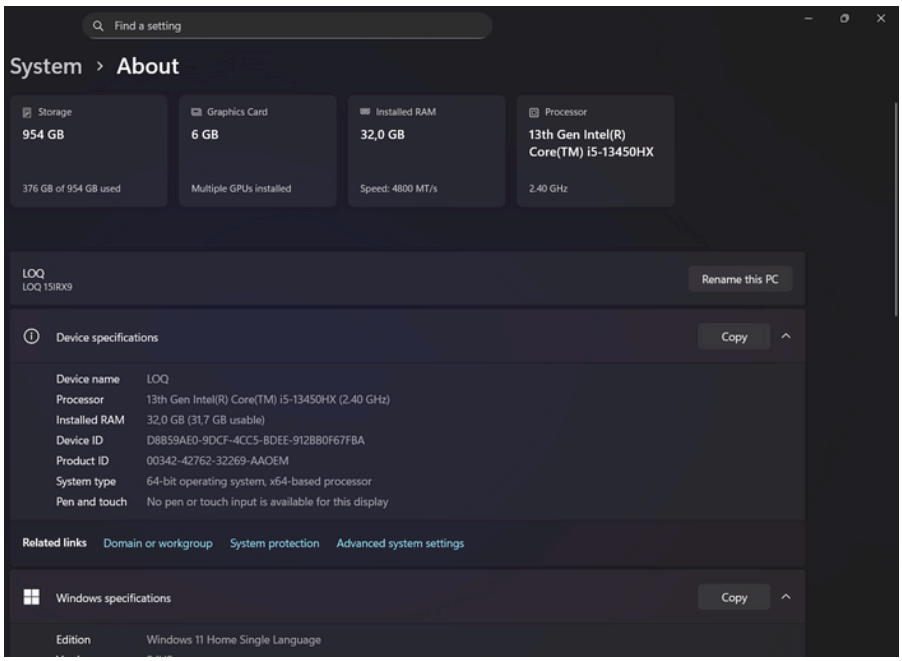
4. R. Penggunaan RAM sebesar 659 MB/ 1973 MB dan Storage 9 GB/ 17 GB



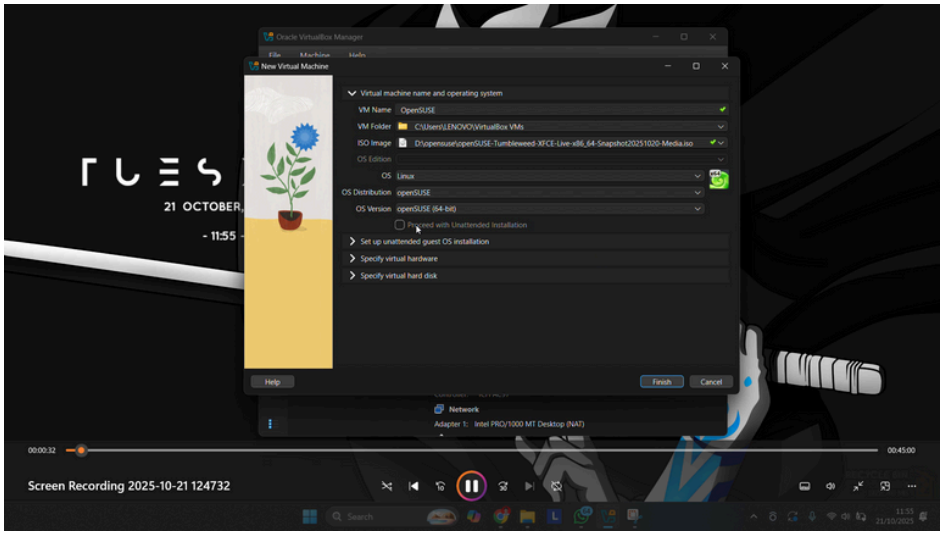
Tahapan Instalasi OpenSUSE OS Tumbleweed
VirtualBox
2GB/10GB

5. Persiapan

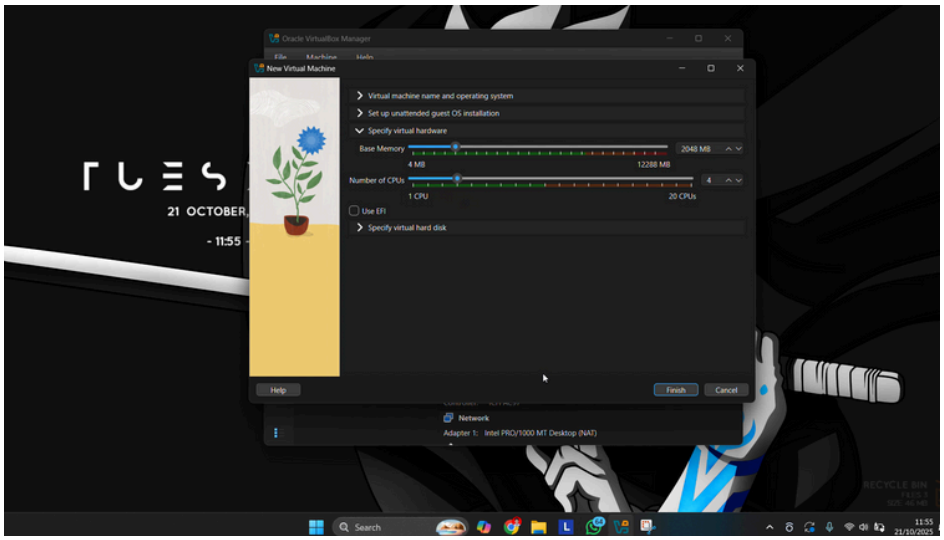
5. A. Spesifikasi laptop/PC memadai (RAM minimal 4 GB, storage kosong 30 GB).



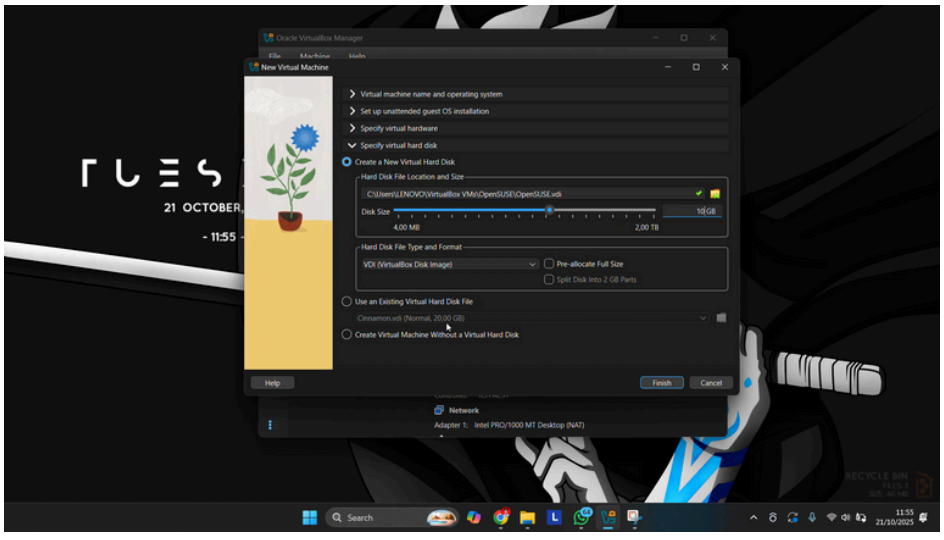
5. B. Masukkan file ISO



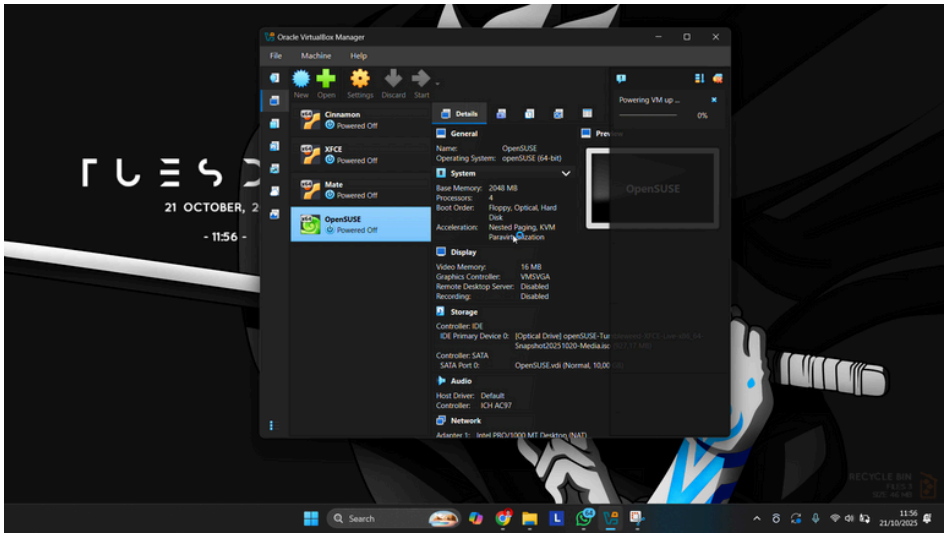
5. C. Atur RAM 2 GB dan 4 CPUs



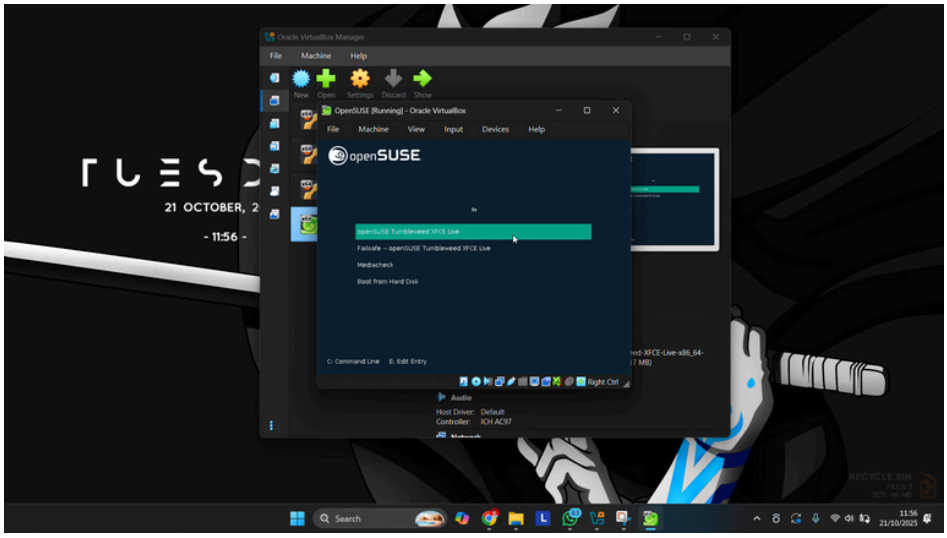
5. D. Atur storage menjadi 10 GB → klik finish



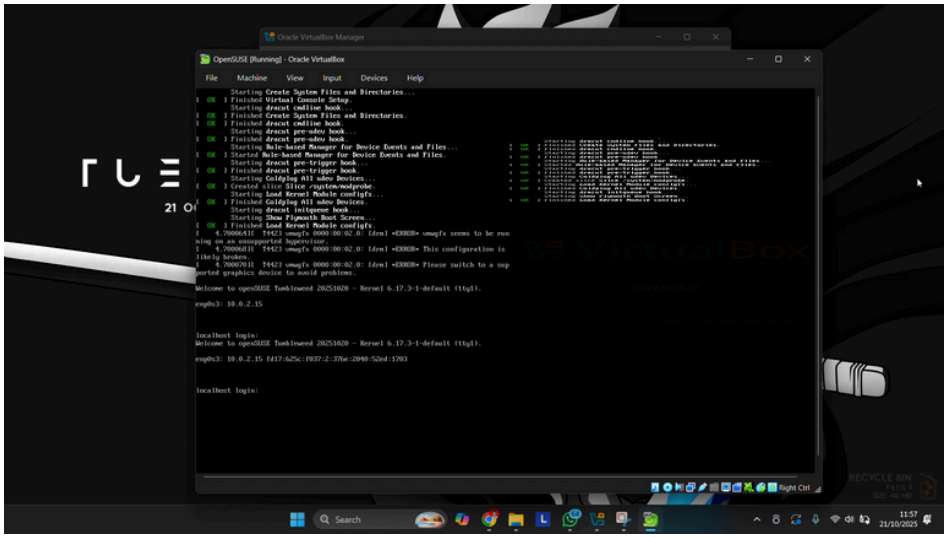
5. E. Klik start → Tunggu proses Powering VM



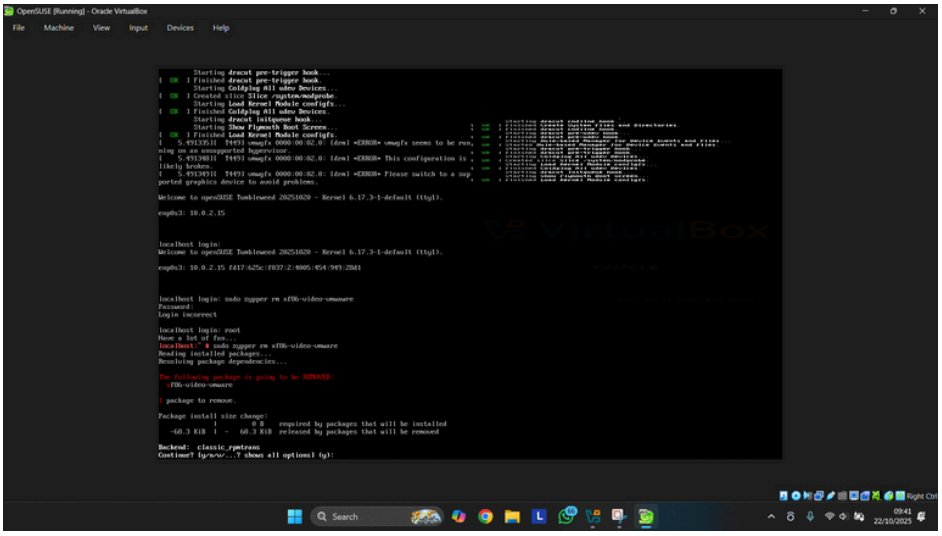
5. F. Klik openSUSE Tumblewood XFCE LIve



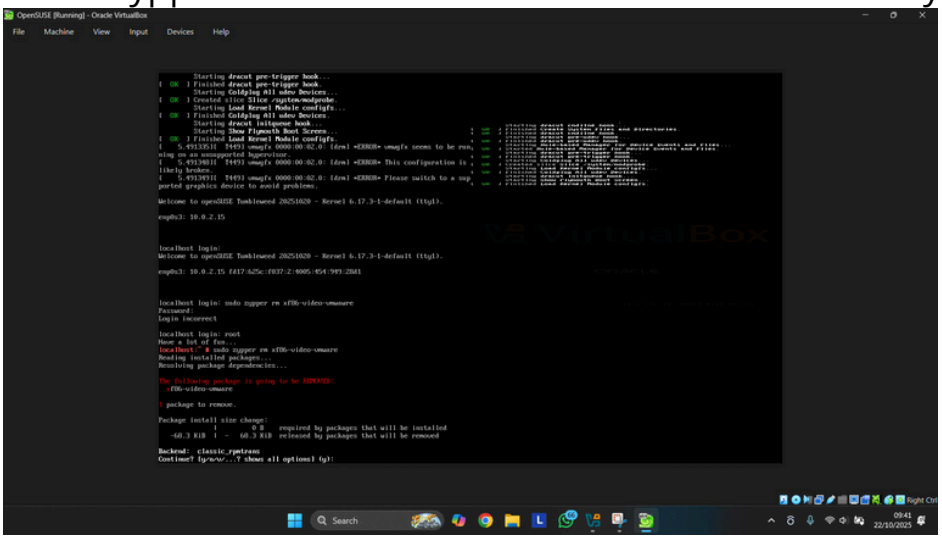
5. G. dikarenakan kendala pada tampilan dekstop maka perlu perbaiki melalui terminal



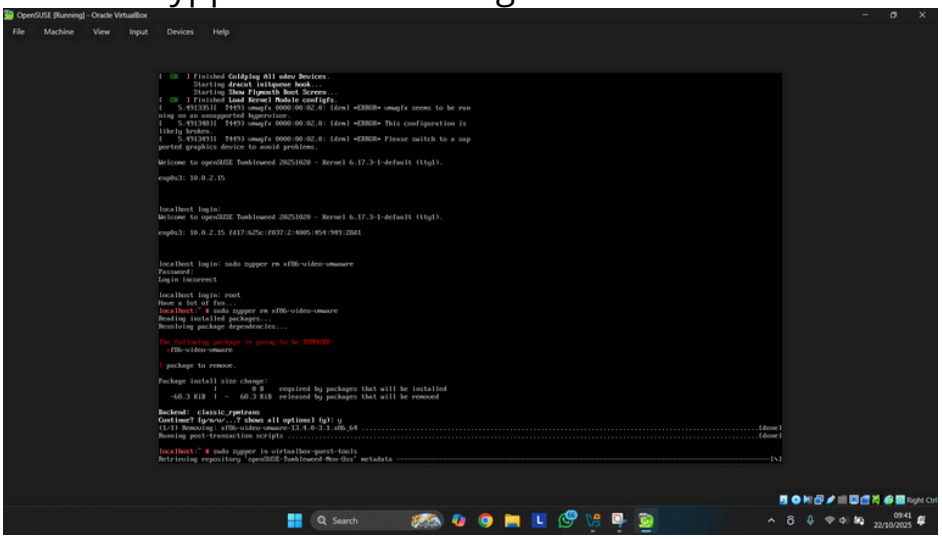
5. H. ketik root untuk login localhost



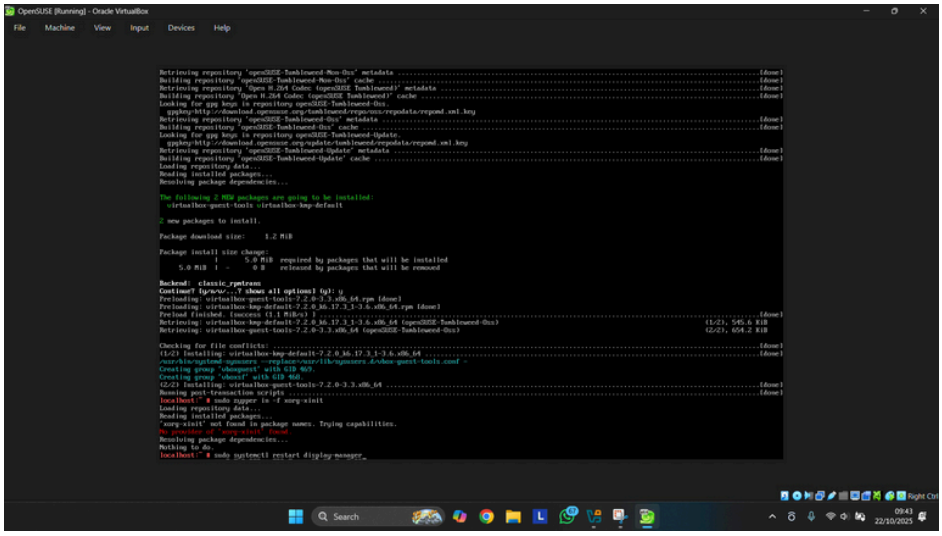
5. I. ketik “sudo zypper rm xf86-video-vmware” enter kemudian “y”



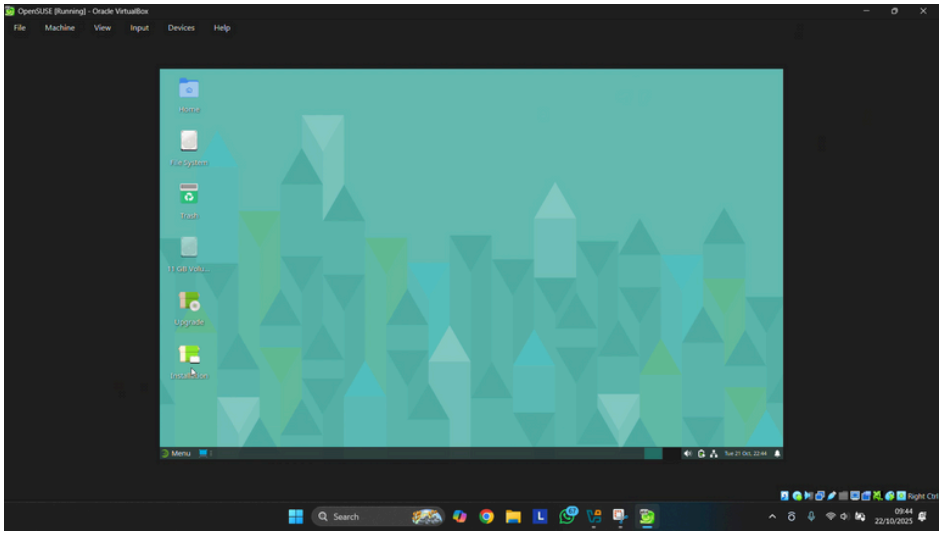
5. J. ketik “sudo zypper in virtualbox-guest-tools”



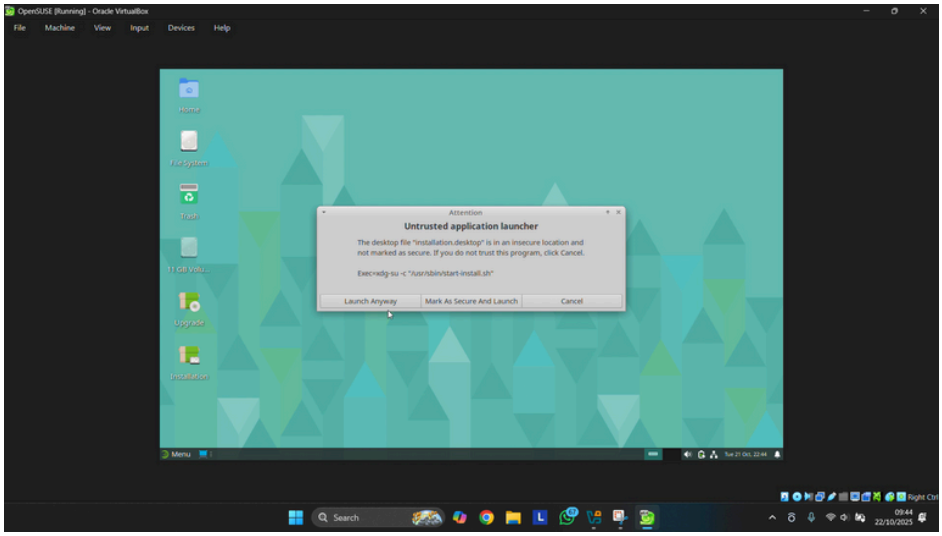
5. K. ketik “sudo zypper restart display-manager”



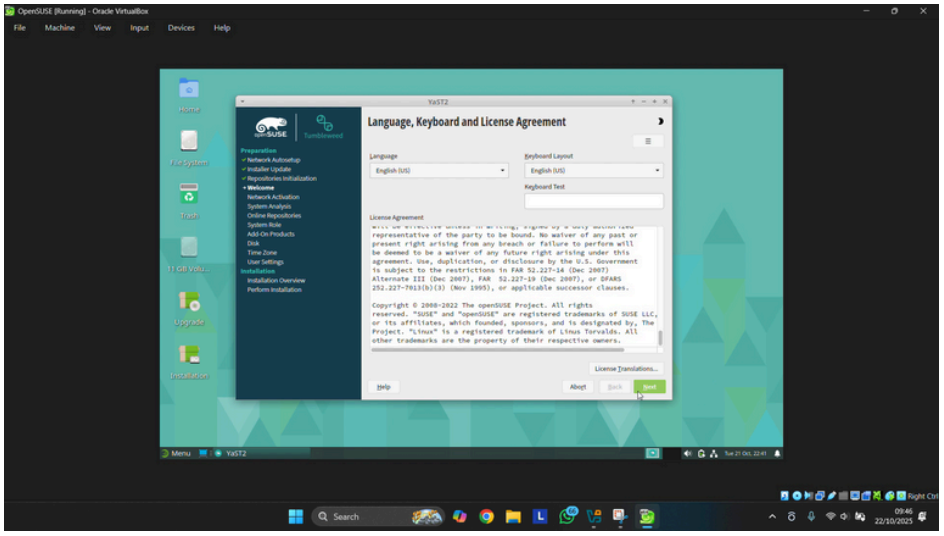
5. L. Tampilan awal OpenSUSE OS



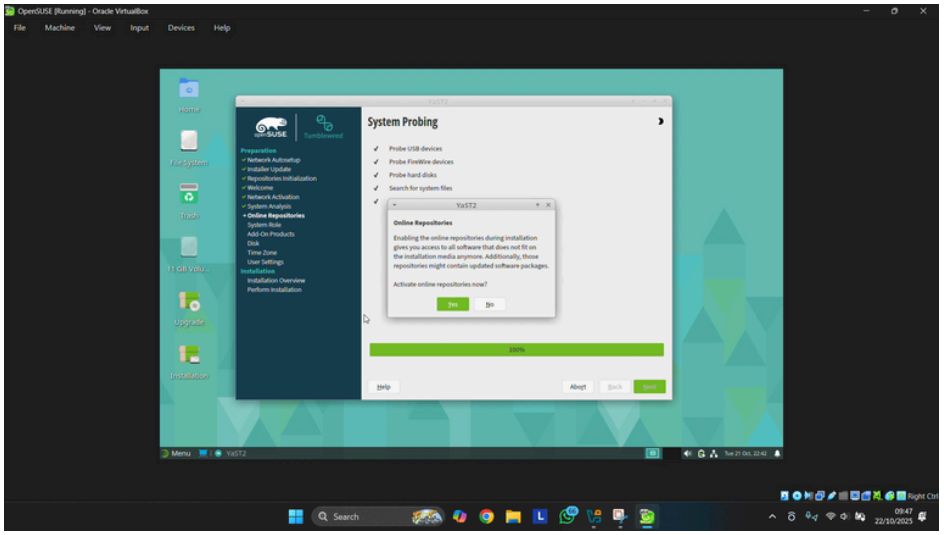
5. M. Klik Installation icon kemudian Launch Anyway



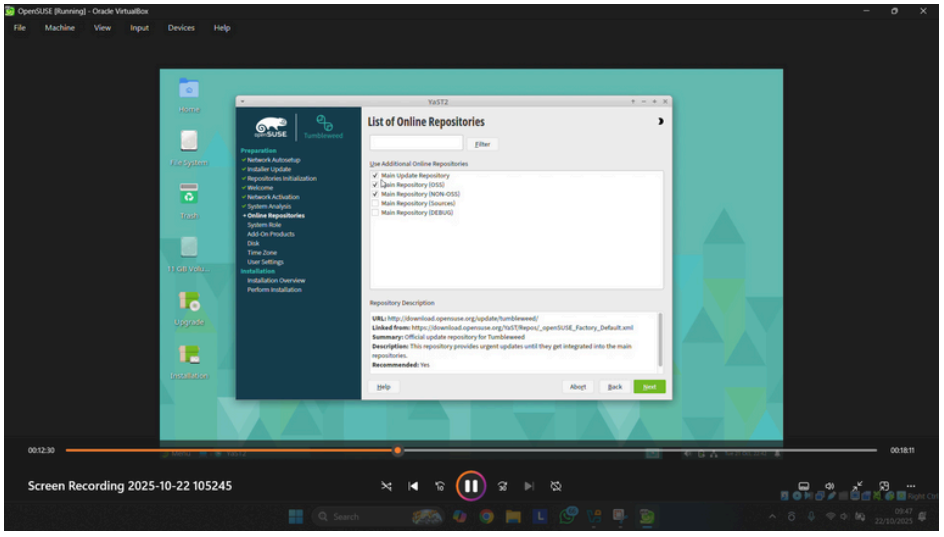
5. N. Atur bahasa dan layout keyboard → Klik next



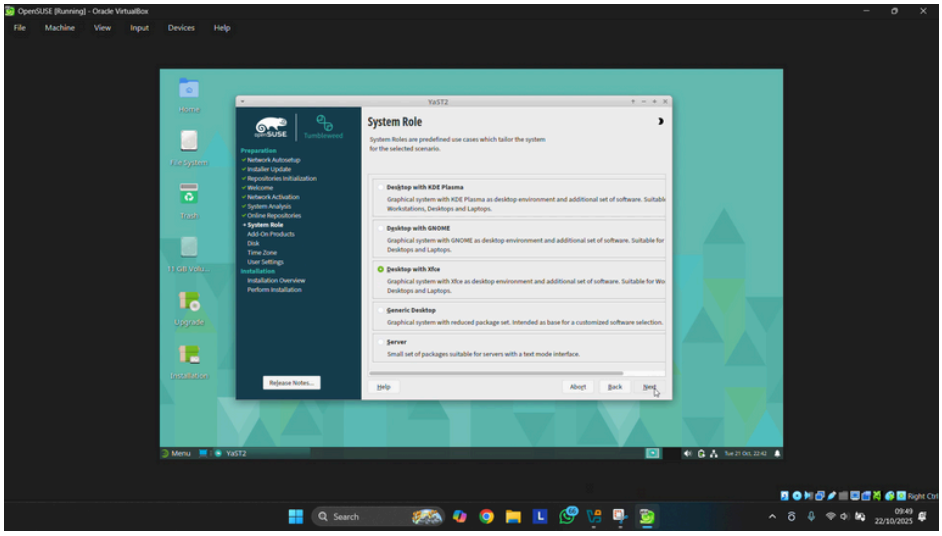
5. O. Untuk instalasi paket yang akan diinstal → Klik next



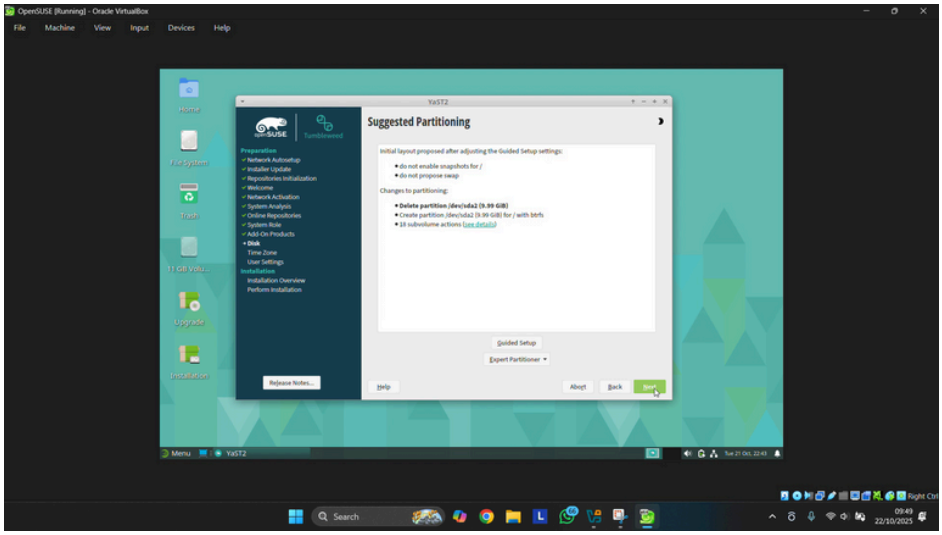
5. P. Pilih server untuk mengunduh paket instalasi yang diperlukan



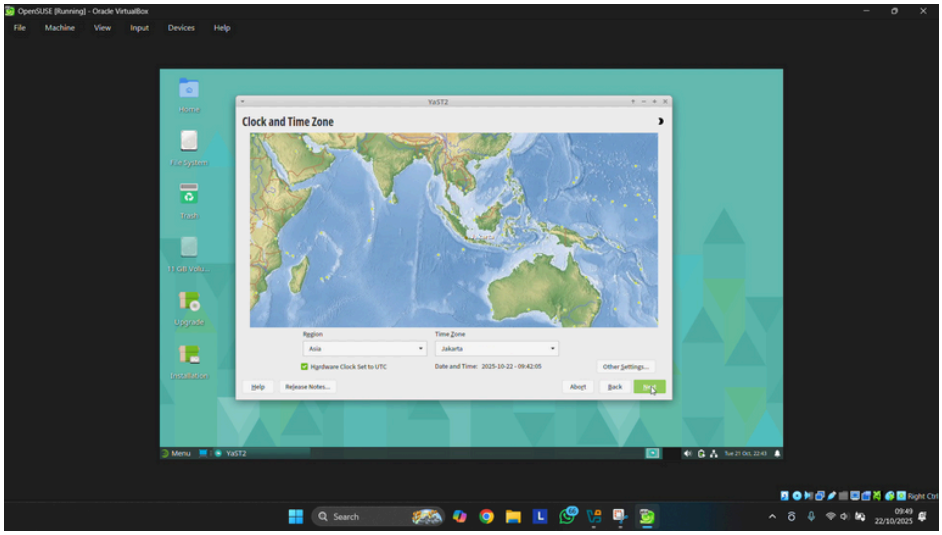
5. Q. pilih tampilan dekstop Xfce



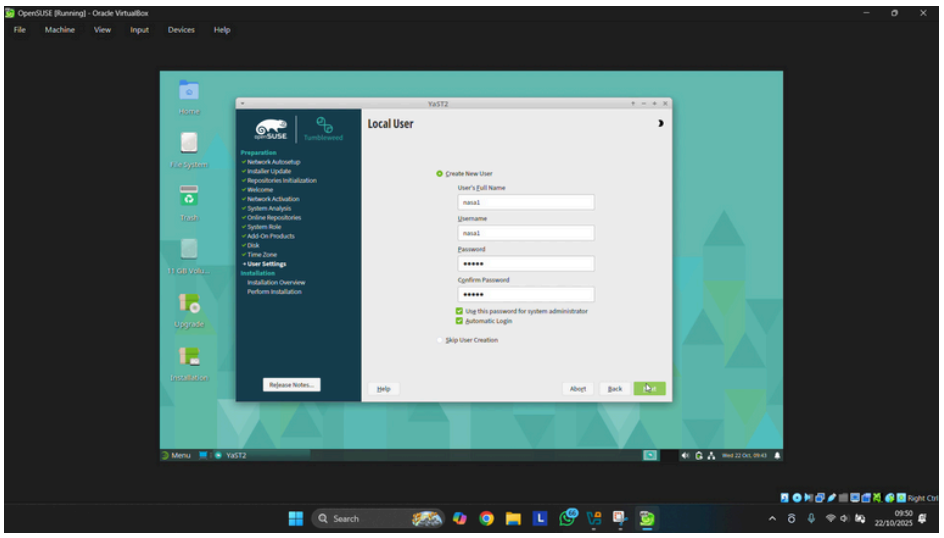
5. R. Atur partisi → Klik next



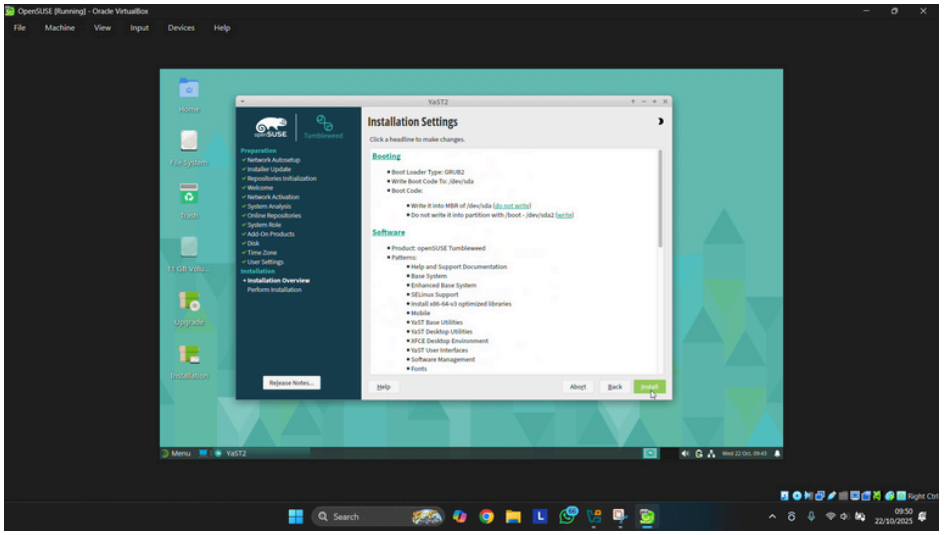
5. S. Pilih wilayah



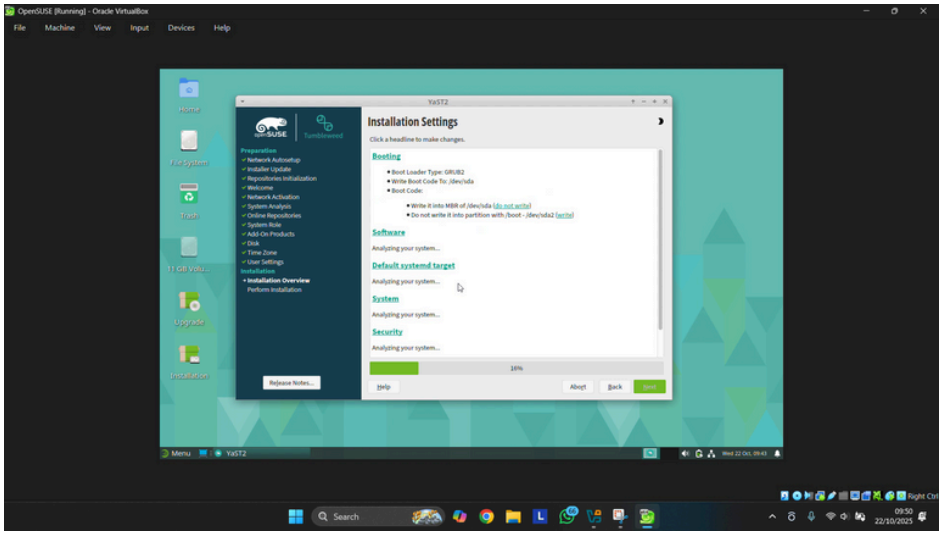
5. T. konfigurasi user dan password untuk login opensuse



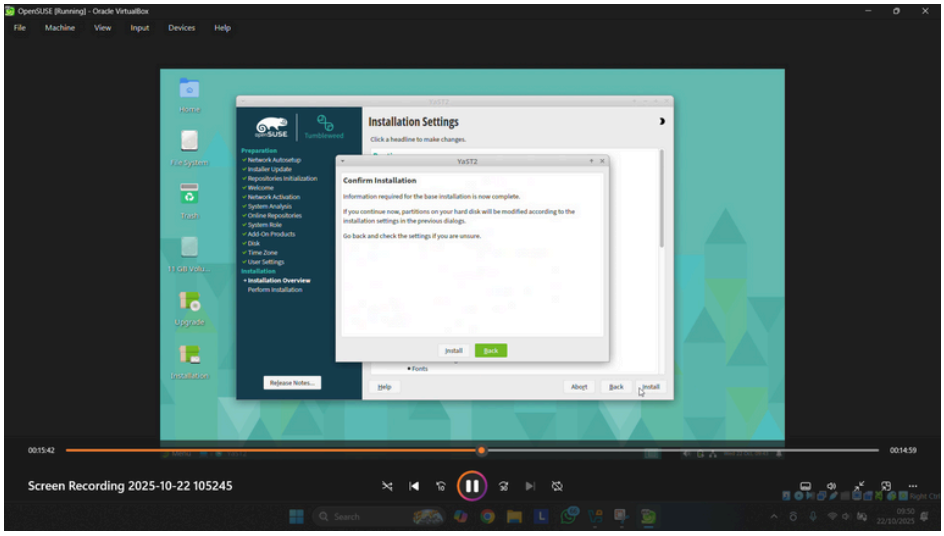
5. U. Cek kembali apakah sudah sesuai → Klik Install



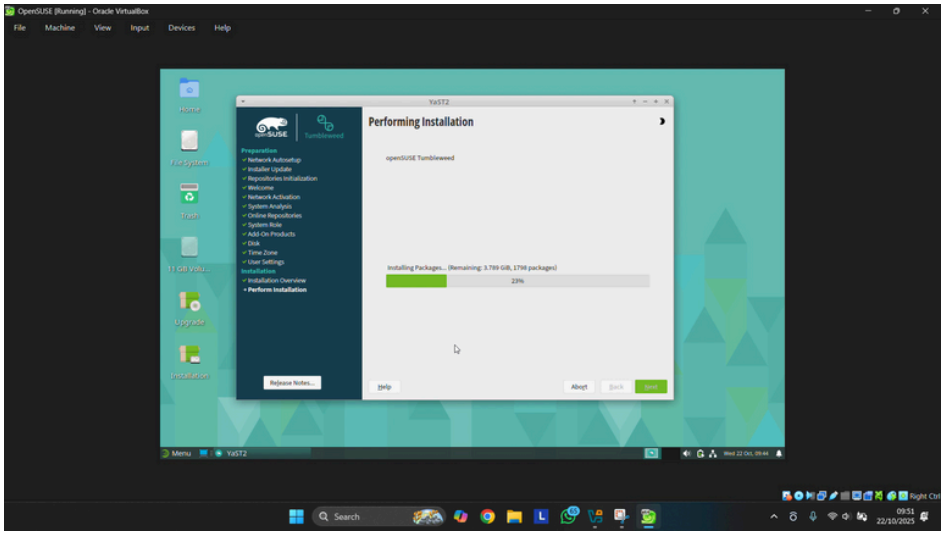
5. V. Tunggu proses analisis OpenSUSE OS



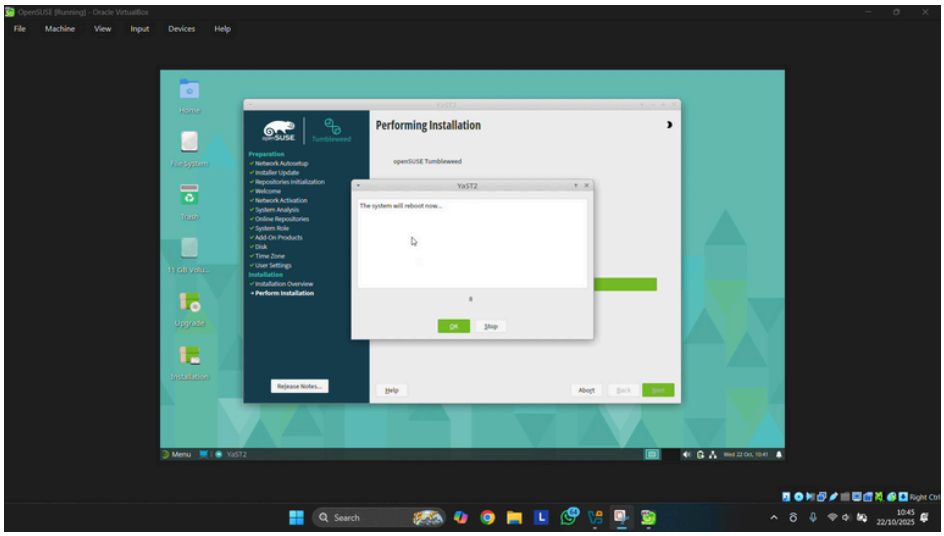
5. W. Lakukan konfirmasi OpenSUSE OS instalasi



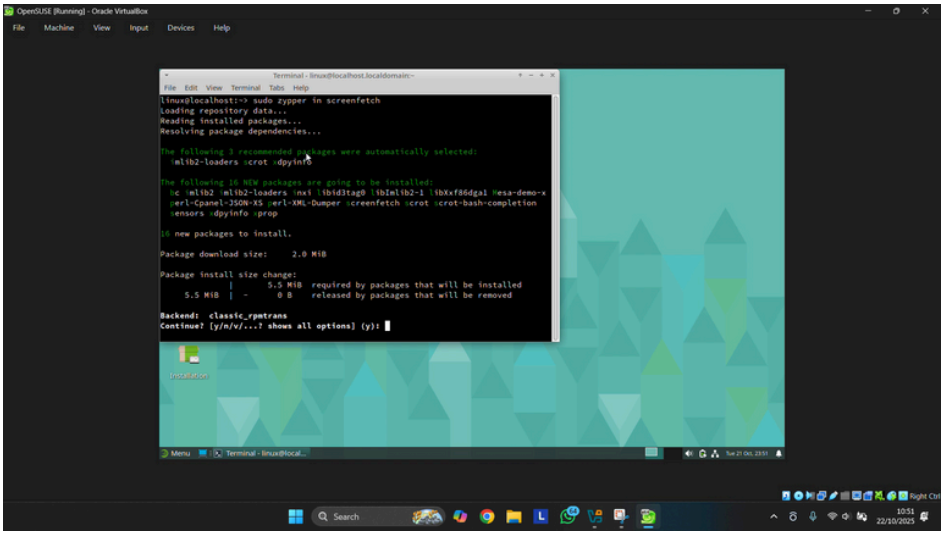
5. X. Tunggu Prosses Install



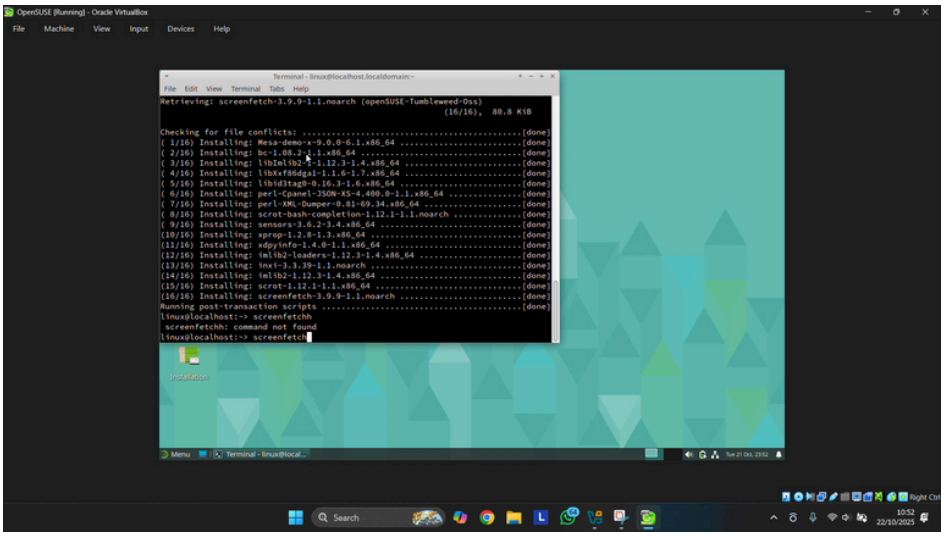
5. Y. Install selesai → klik ok



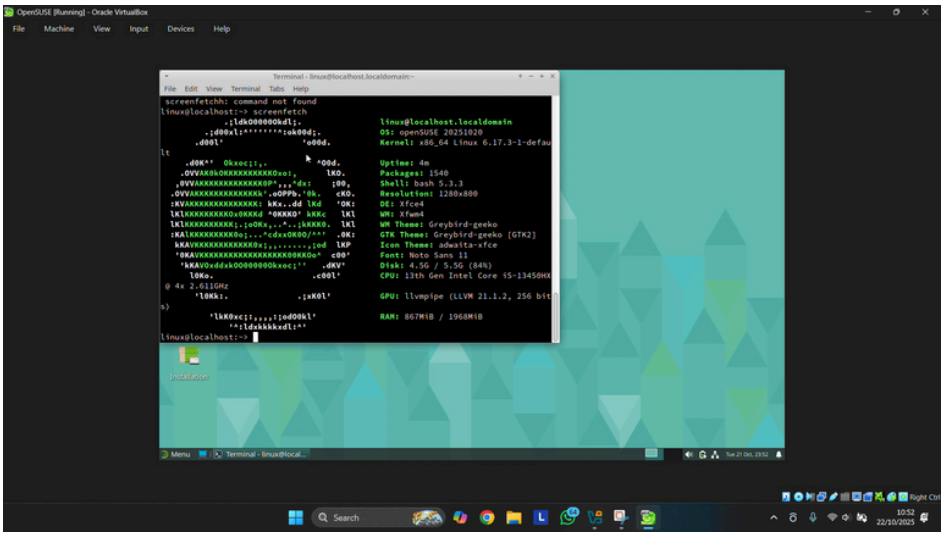
5. Z. Buka terminal → ketik “Sudo zypper in screenfetch”



5. AA. Ketik “screenfetch”



5. AB. Spesifikasi RAM dan DISK



Tabel Debian OS dan OpenSUSE

Aspek	Debian 13.1.0	OpenSUSE Tumbleweed
Memory RAM	659MB/ 1973MB	867MB/ 1968MB
Storage DISK	9GB/20GB	4,5GB/10GB
Model rilis	Rilis stabil dengan titik rilis (point release)	Model rolling release - terus diperbarui setiap saat tanpa “versi besar” reguler
Versi XFCE	Xfce 4.20 sebagai salah satu pilihan DE.	Xfce sebagai pilihan DE, dengan paket-Xfce yang juga cukup terkini.
Stabilitas Update	Lebih fokus ke stabilitas, kompatibilitas jangka panjang	Fokus ke fitur terbaru, driver terbaru, paket terbaru
Pengelola Paket	Paket berbasis APT (.deb) dengan repos Debian Stable	Paket berbasis RPM/Zypper + YaST
Resiko	Lebih rendah risiko “rusak” saat update besar karena model rilis stabil dan jarang perubahan besar	Risiko sedikit lebih tinggi karena perubahan terus-menerus; meskipun openSUSE Tumbleweed dikenal cukup teruji
File ISO desktop	Banyak web mirror untuk mengunduh versi live/installer Debian 13 dengan Xfce 4.20 langsung	Tipe Snapshot Backup ISO di Tumbleweed tersedia dan bisa dipilih/di-install DE tertentu; dokumentasi untuk Xfce juga tersedia

Troubleshooting Praktikum

Praktikum instalasi debian cukup lancar dan minim kendala. Terdapat spesifikasi disk khusus diatas 10GB agar dapat terinstall di VirtualBox. Banyak pepatah berkata Website debian adalah seleksi bagi calon user. Dan kami dapat menjawabnya :
“Nah... Well Played. We still win”

Kami menemukan sedikit kendala pada file ISO OpenSUSE Tumbleweed, dimana tampilan desktop perlu di fix manual oleh kelompok. Sistem di VirtualBox failed untuk memuat GUI Desktop. Pada proses instalasi juga sedikit lama dikarenakan mendownload paket tambahan secara default. Pada akhirnya praktikum berjalan lancar dan dapat diselesaikan secara optimal.

Berikut link rekaman praktikum kelompok 2:

https://drive.google.com/drive/u/3/folders/1moaPB1tQqtV7PQ5yEJT92Bc6vZyboZ6R

Tidak ada filosofi resmi di balik logo Debian, namun ada spekulasi bahwa logo tersebut melambangkan "asap ajaib" yang membuat komputer bekerja. Logo ini tidak memiliki filosofi atau makna resmi yang diumumkan oleh proyek Debian itu sendiri.

Logo OpenSUSE, yang berbentuk gelembung spiral hijau, tidak memiliki filosofi khusus yang terperinci dalam dokumen resmi yang tersedia, namun dapat diartikan sebagai simbol dari sistem operasi sumber terbuka yang stabil, aman, dan mudah digunakan.

LAPORAN PRAKTIKUM INSTALASI Debian OS dan OpenSUSE di VirtualBox

Disusun oleh:
Kelompok : 2

Anggota Tim :

a. Bima Adhi Pratama Kharis
b. Alif Satya Wicaksono
c. zakka noviatama
d. MUHAMMAD DZAKY ALTHAF

Mata Kuliah : Sistem Operasi

Dosen : Yahya Nur Ifriza, S.Pd., M.Kom.